this could significantly reduce reliance on TypeScript in the future.

While advancements in JavaScript are promising, it is unlikely that TypeScript will become obsolete in the near future. TypeScript offers advanced features like interfaces, enums, and sophisticated type inference that go beyond basic type annotations. It also enjoys strong ecosystem support, with many frameworks and libraries, such as Angular, treating it as a first-class citizen, making it integral to modern development workflows. Additionally, TypeScript provides tooling and developer experience through features like autocompletion and inline documentation in IDEs, significantly enhancing productivity.

This combination of technical expertise and industry knowledge highlights a candidate's ability to solve problems effectively and adapt to evolving technologies. By showcasing this unique blend of skills, candidates like Jane can differentiate themselves from others and demonstrate their value as innovative, forward-thinking developers.

In the next chapter, we will focus on React’s core concepts to explore the fundamentals essential to modern web frameworks and libraries, including declarative UI, component-based architecture, state management, and more.

3.12 Summary

- Be prepared to explain how JavaScript manages execution contexts and scope (global, function, block), and how the scope chain controls variable access.
- Demonstrate a solid grasp of functional programming concepts such as pure functions, immutability, and higher-order functions to write cleaner, more maintainable code.
- Clearly articulate how the event loop, promises, and `async`/`await` enable non-blocking, event-driven programming in JavaScript.
- Show understanding of garbage collection, memory leak prevention, and best practices for organizing modular code.
- Highlight the advantages of TypeScript’s static typing, early error detection, and enhanced developer experience, which drive its widespread adoption.